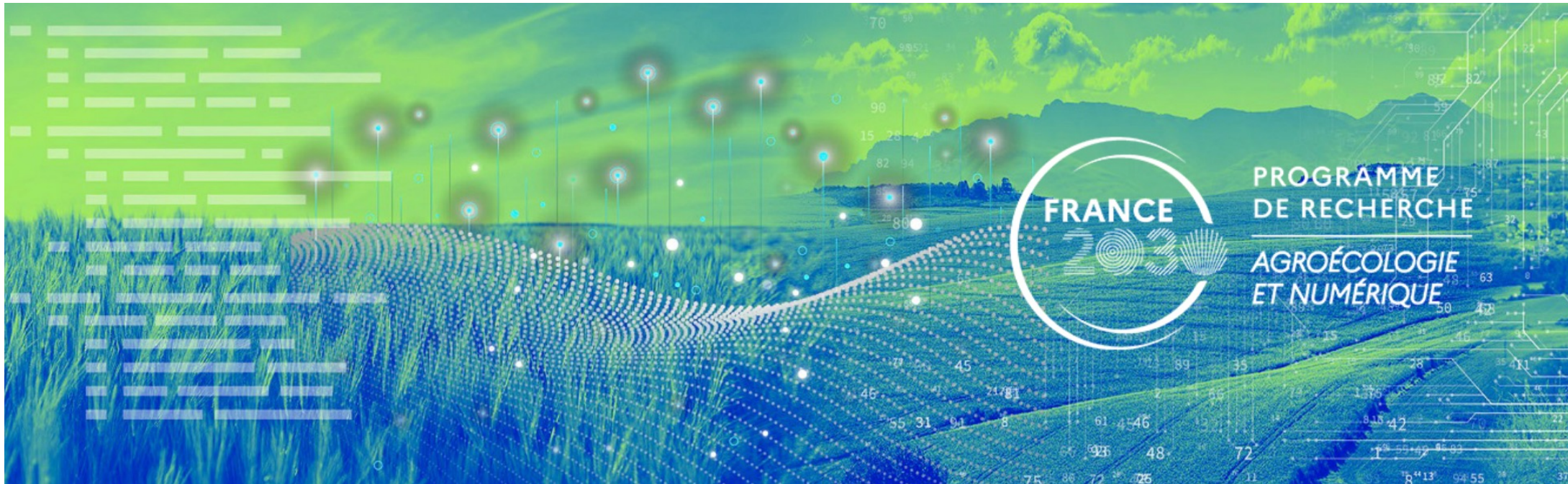




# Building the LoRa IoT platform

## Part 2: the customized IoT gateway

Prof. Congduc Pham  
<http://www.univ-pau.fr/~cpham>  
 Université de Pau, France

Congduc.Pham@univ-pau.fr



Wireless Sensors Made Simple  
 for agroecology & sustainable agriculture

Powered by technologies developed in  Intel-IrriS



# The customized IoT gateway

- ⦿ The customized IoT gateway is based on the generic WaziGate
- ⦿ WaziGate is an IoT LoRa Gateway developed by WAZIUP
- ⦿ WaziGate implements the edge-enabled IoT gateway approach
  - ⦿ customized applications can be directly hosted in the gateway
  - ⦿ the gateway can easily work without Internet connectivity
  - ⦿ data are available to end-users in an embedded database
  - ⦿ web-based visualization module provides graphical user interface
- ⦿ You can find all the WaziGate documentation on the [WaziGate documentation page](#).
- ⦿ The customized IoT gateway adds many more features
  - ⦿ Simple configuration for all supported sensor types
  - ⦿ Sensor dashboard based on Home Assistant
  - ⦿ Automatic backup and simple restore procedure for sensor data
  - ⦿ Display on OLED screen, ...



# Install software for your gateway

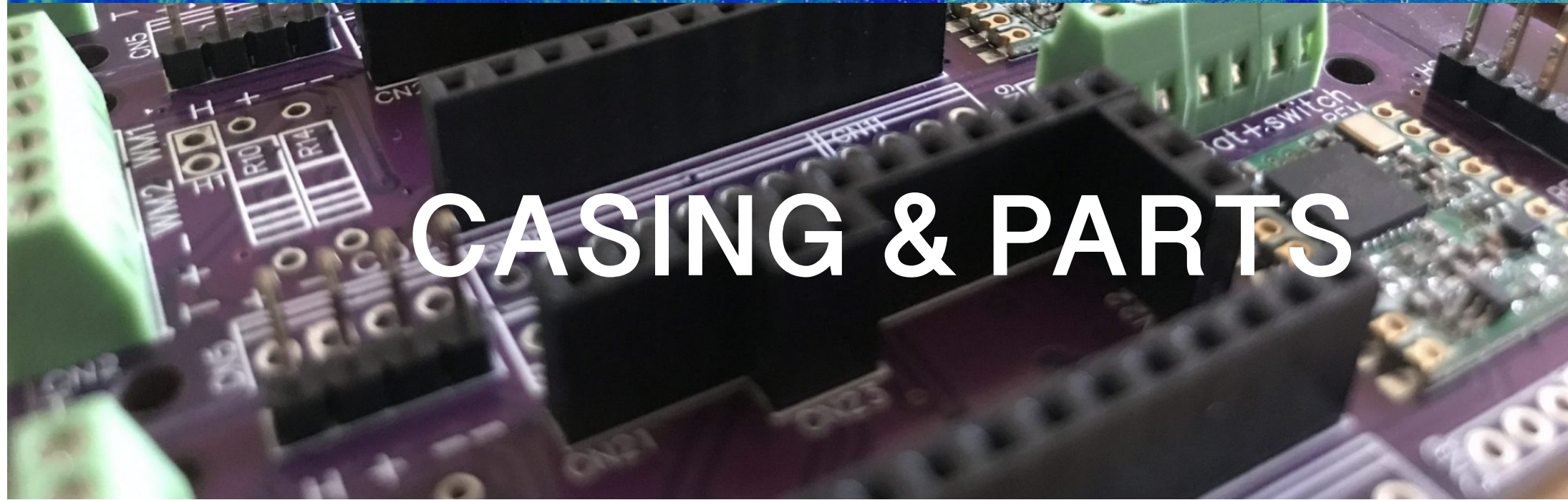
## ● **PEPR AgriFutur gateway distribution (RPI3B/3B+/4B)**

- comes pre-configured
- will work out-of-the box with the default sensor devices
- Including the LoRaCAM-AI device
- **Download the gateway SD card image from project GitHub**
- [https://github.com/CongducPham/PEPR\\_AgriFutur/tree/main/Gateway](https://github.com/CongducPham/PEPR_AgriFutur/tree/main/Gateway)
- Image uses EU868 frequency band
- EU433 or other bands can be configured afterwards
- Flash the SD card image on a class 10 SD card



Look at the video produced for INTEL-IRRIS for the main steps  
Video at t=124s: <https://youtu.be/j-1Nk0tv0xM?t=124>









# Raspberry PI enclosure

- You can take a simple commercial plastic RPI case to easily drill holes for the antenna connector, or use a 3D printed enclosure

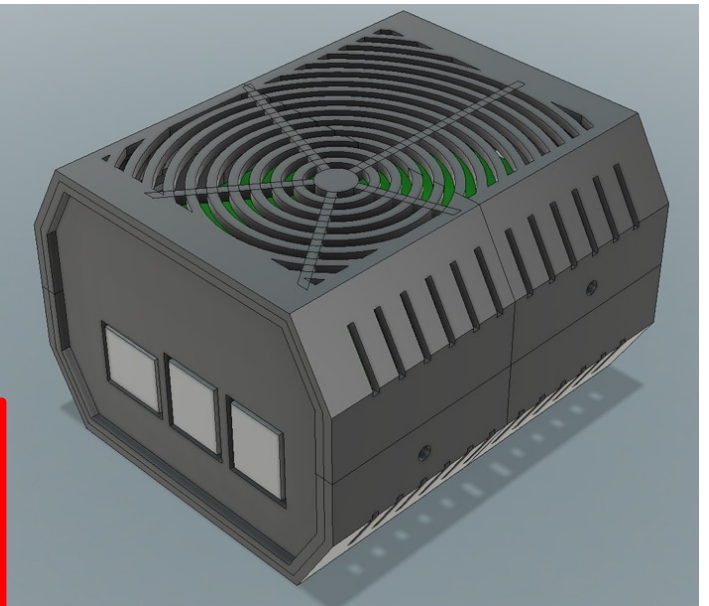
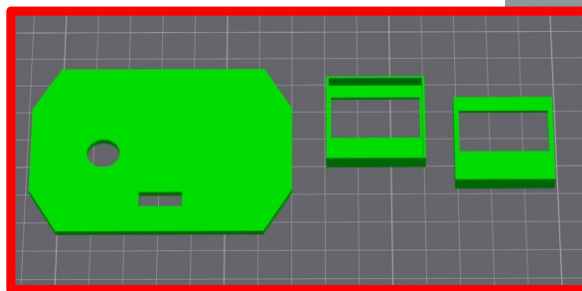
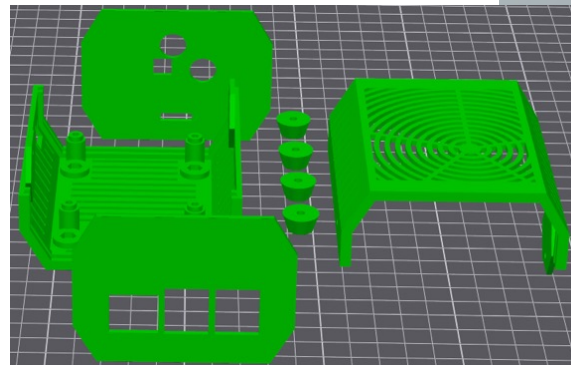


RPI3B/3B+



RPI4B

3D printed from <https://www.thingiverse.com/thing:3012012>



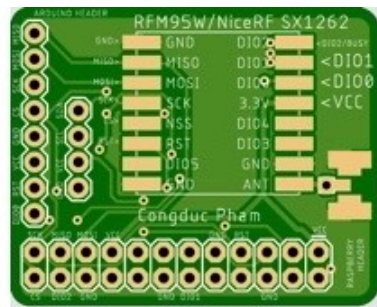
We provide modified & additional parts on GitHub



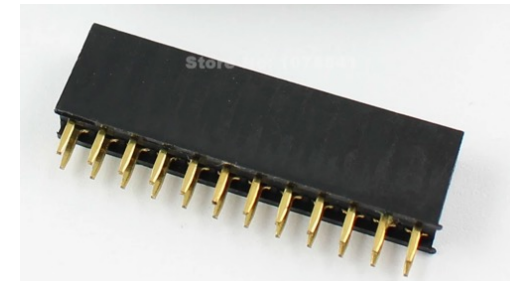
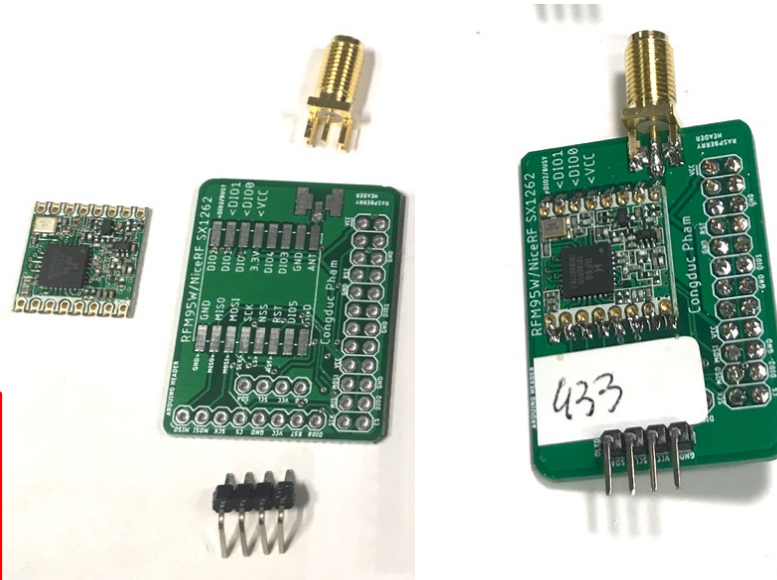
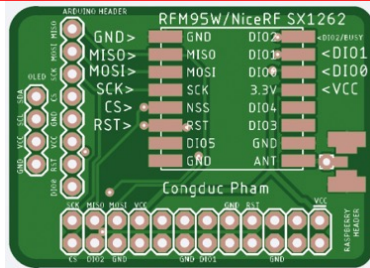
# Single-channel: the LoRa radio hat

Information on how to manufacture and order a PCB:

<https://github.com/CongducPham/PRIMA-Intel-IrriS/blob/main/Tutorials/Intel-Irris-PCB-update-PCBA.pdf>



RFM95W/NiceRF SX1262  
breakout



2-row 12-pin header  
or  
2 X 1-row 12-pin header

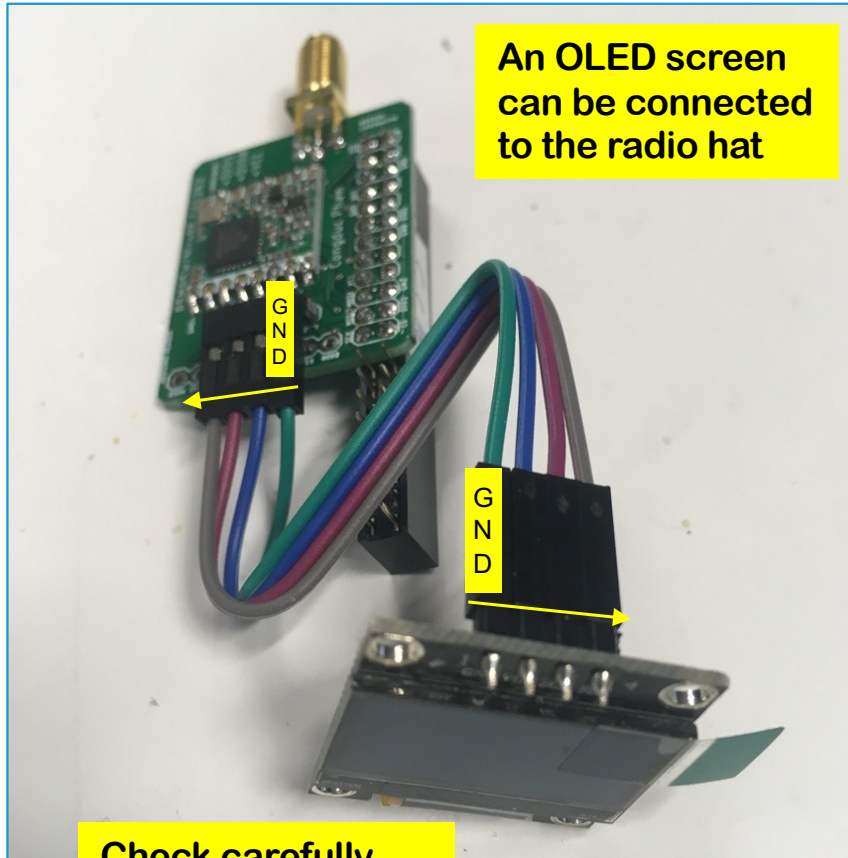


Get the zipped Gerber file from:

[https://github.com/CongducPham/PEPR\\_AgriFutur/blob/main/PCBs/RFM95Breakout\\_2020-11-13-long.zip](https://github.com/CongducPham/PEPR_AgriFutur/blob/main/PCBs/RFM95Breakout_2020-11-13-long.zip)



# Installing OLED



Check carefully  
the pin order  
which is reversed



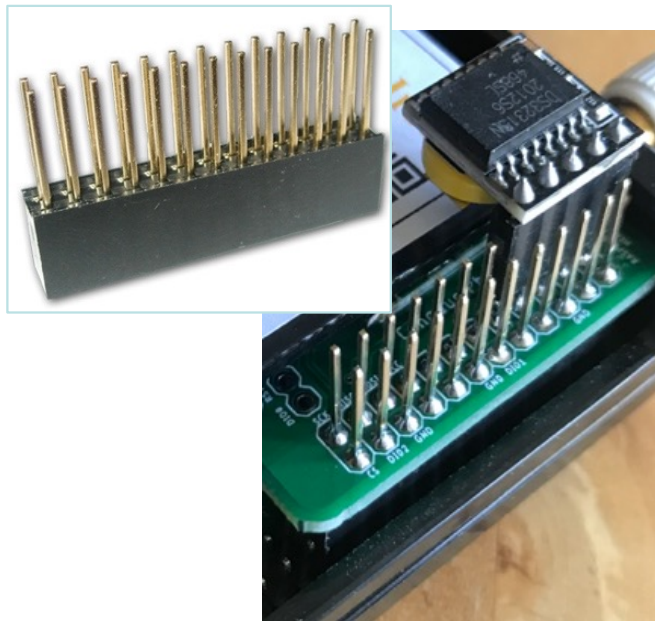




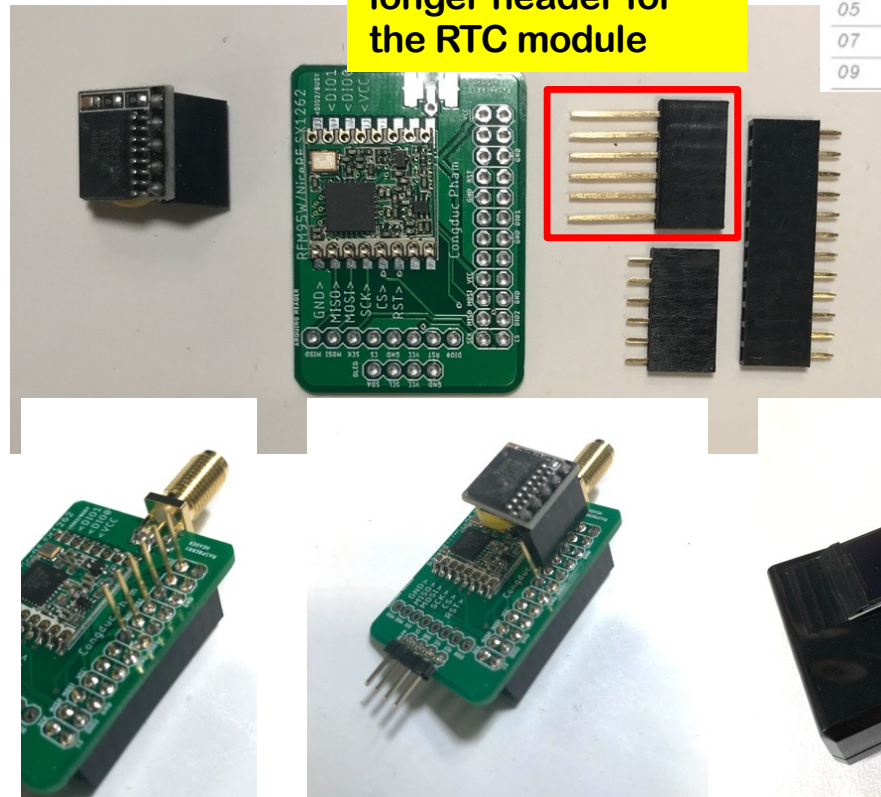
# Connecting an RTC module

Use a DS3231

- Several possibilities depending on availability of components



2/ use only a 6-pin longer header for the RTC module



| Pin# | NAME                             |
|------|----------------------------------|
| 01   | 3.3v DC Power                    |
| 03   | GPIO02 (SDA1 , I <sup>2</sup> C) |
| 05   | GPIO03 (SCL1 , I <sup>2</sup> C) |
| 07   | GPIO04 (GPIO_GCLK)               |
| 09   | Ground                           |

1/ Solder a header with longer pins on the radio hat. You can then cut the remaining pins if you want





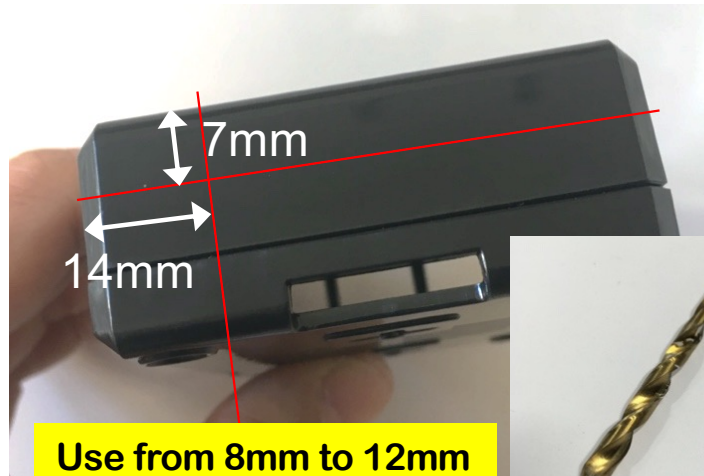
# Preparing the enclosure

- Cost-effective RPI model is RPI3B/3B+

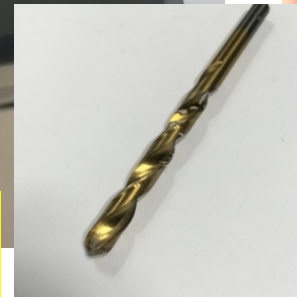
- plug in the radio LoRa hat
- The OLED screen is optional but highly recommended



- Drill a hole for the SMA connector on the RPI enclosure, at the SD card slot side
- <https://fr.aliexpress.com/item/32718435597.html>



Use from 8mm to 12mm  
drill bit for metal

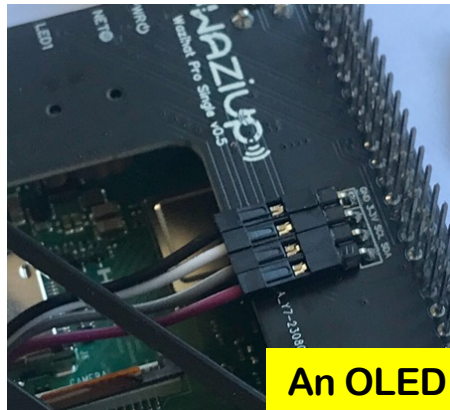




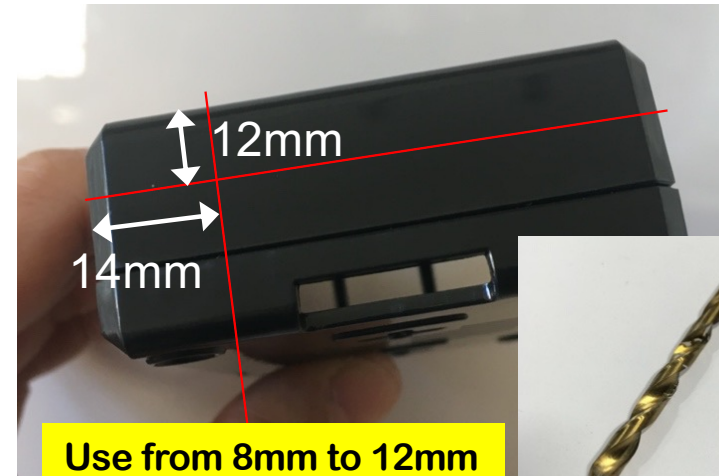
# Using WAZIUP LoRa gateway hat



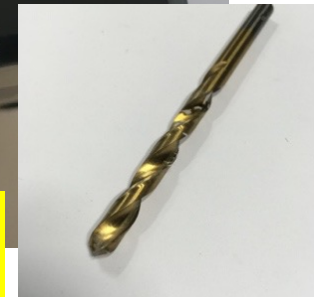
LoRa gateway hat  
with on-board RTC,  
no need for external  
RTC module



An OLED screen can  
directly be connected  
to the radio hat



Use from 8mm to 12mm  
drill bit for metal

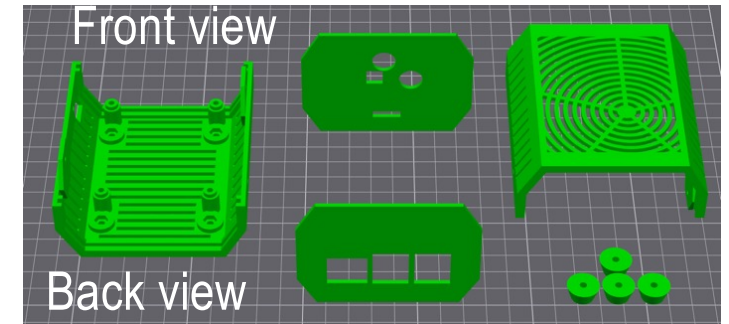






# Using the Thingiverse case

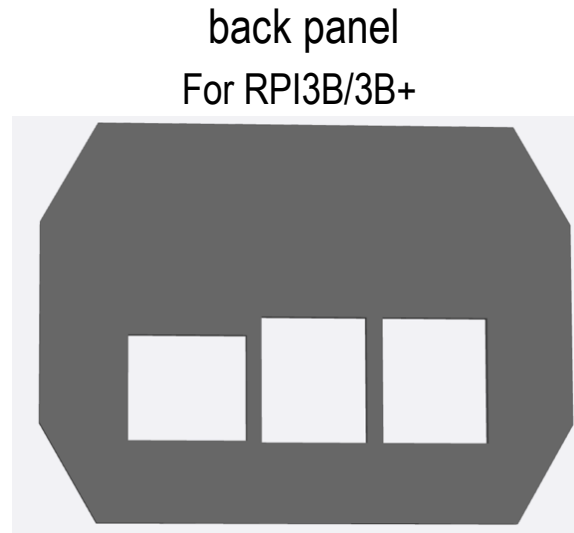
<https://www.thingiverse.com/thing:3012012>



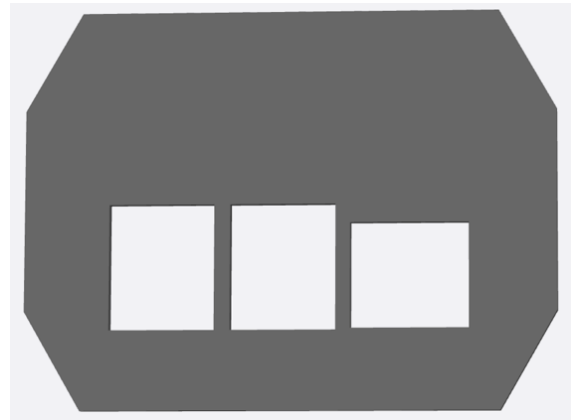
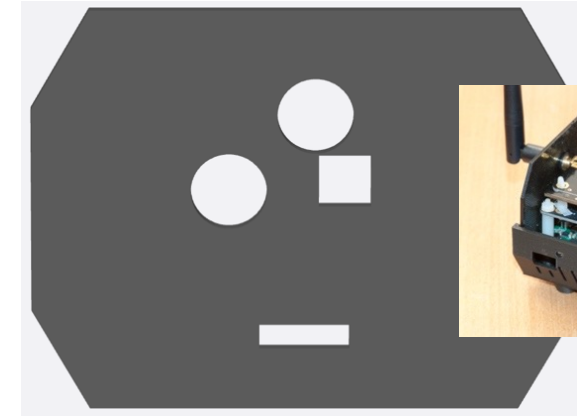
Front view: initial front panel designed for RAK831



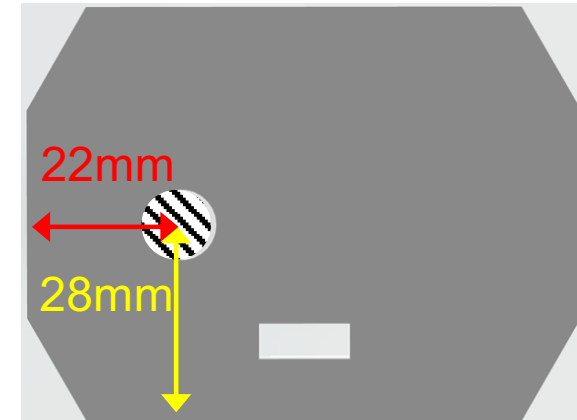
RPI3B/3B+



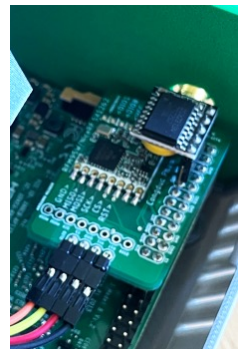
back panel  
For RPI3B/3B+



Simply flip the back panel for RPI4B

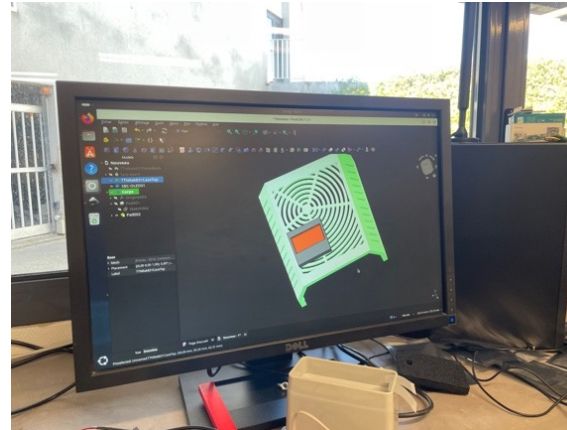
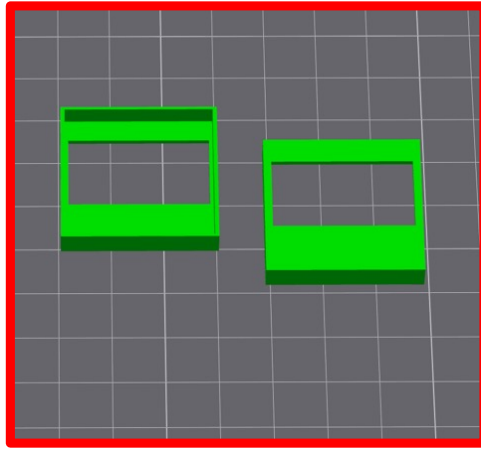


We provide a modified front panel





# OLED cover with the Thingiverse case



6cm x 8cm  
sticker







# Multi-channel/Multi-SF

- It is possible to have a multi-channel/multi-SF gateway by connecting a LoRa radio concentrator such as the RAK2245 or the new RAK5146 with the RAK5146 Pi Hat (right photo)
- The LoRa radio concentrator will be automatically detected
- IMPORTANT:** a reliable 5V 2.5A power source is necessary



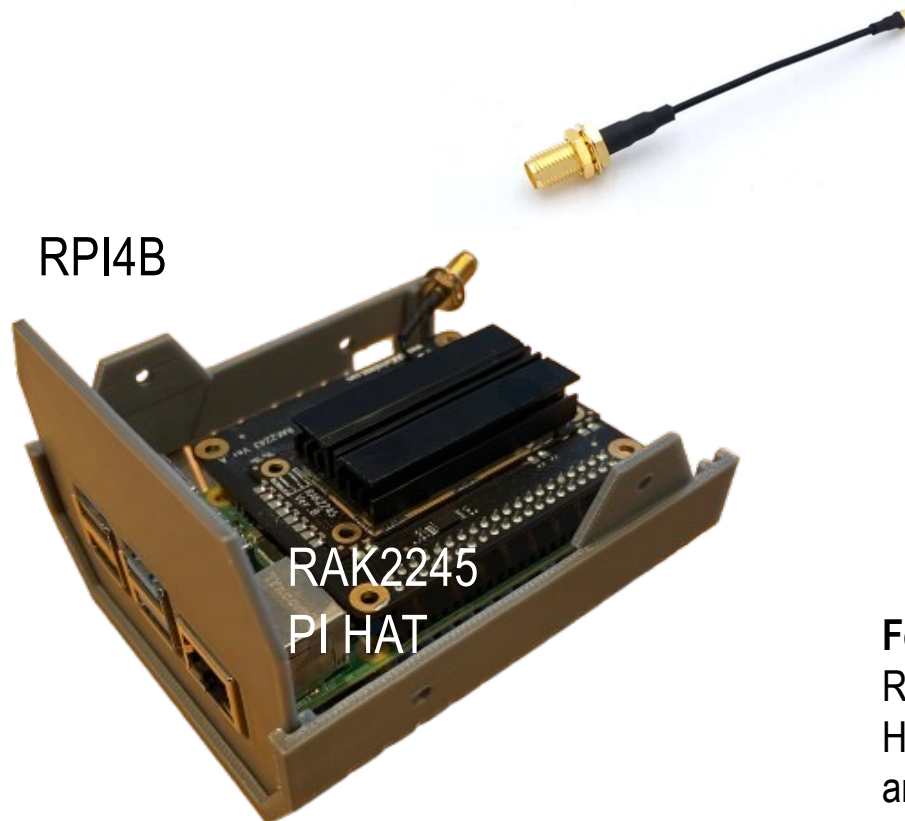
RAK2245



RAK5146



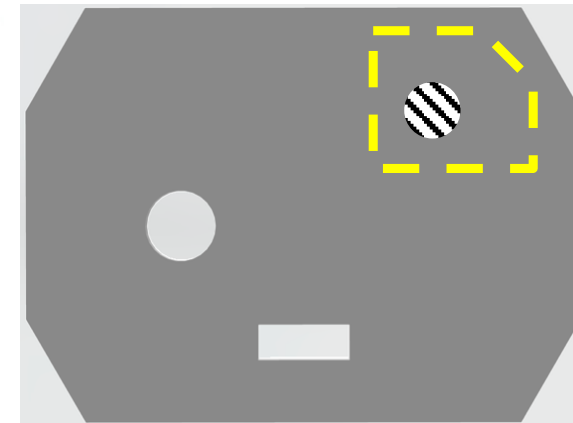
# Thingiverse enclosure with multi-channel



RPi4B

RAK2245  
PI HAT

Front view



**For the multi-channel radio concentrator** such as the RAK2245 or the new RAK5146 with the RAK5146 Pi Hat, simply drill a 8mm hole somewhere in the top-right area of the front panel for the ufl-SMA pigtail





# Boot with mSATA SSD

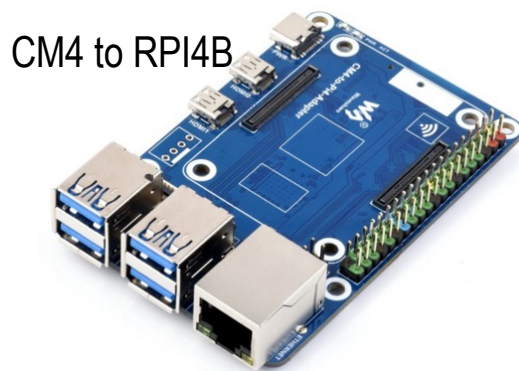
- For more reliability, the SD card image can be flashed on an mSATA SSD disk (with an mSATA-USB adapter) to avoid SD card issues
- 16GB is sufficient





# Compute Module 4 with eMMC

- Or, use a Compute Module 4 with an adapter board to have an RPI 3B/3B+/4B based on eMMC to avoid SD card issues

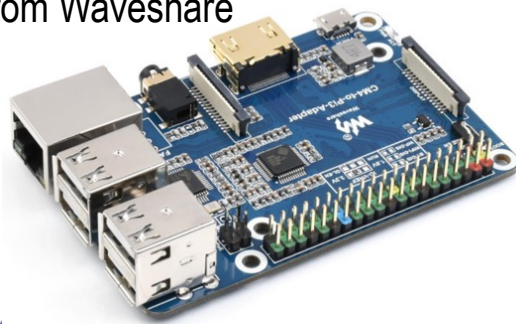


CM4 to RPI4B



CM4

RPI adapter boards  
from Waveshare



CM4 to RPI3B/3B+



Mini Base A with aluminium casing



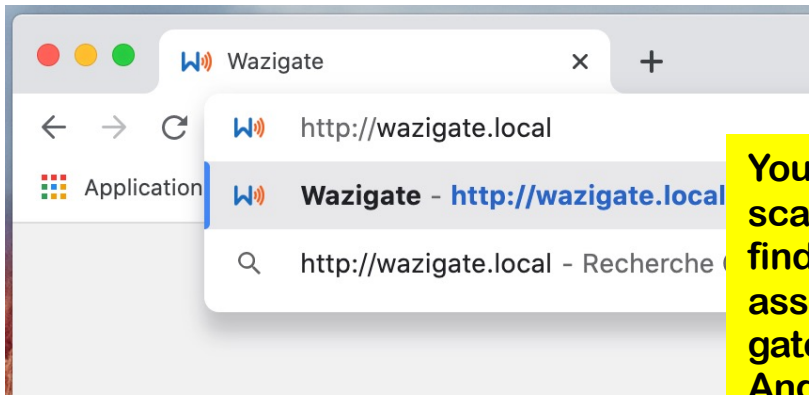






# Checking the gateway (Ethernet cable)

- ⦿ Connect the gateway to your laptop which has Internet
- ⦿ Enable Internet sharing, laptop provides IP address to gateway
- ⦿ Power the gateway, wait 3-4mins for boot process
- ⦿ Open web navigator. Go to <http://wazigate.local> or use IP address



**You can use an IP scanner program to find the IP address assigned to your gateway (such as Angry IP Scanner)**

**Wazigate** Login to the Wazigate dashboard

Username\*  
admin

Password\*  
.....

**LOGIN**

Default username: **admin**  
Default password: **loragateway**

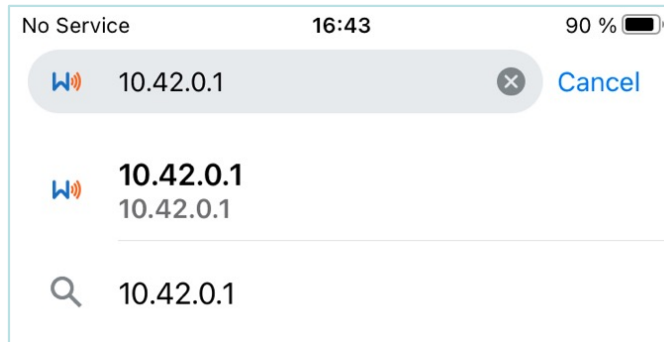
- ⦿ Use default login to connect
  - ⦿ User: admin
  - ⦿ Password: loragateway



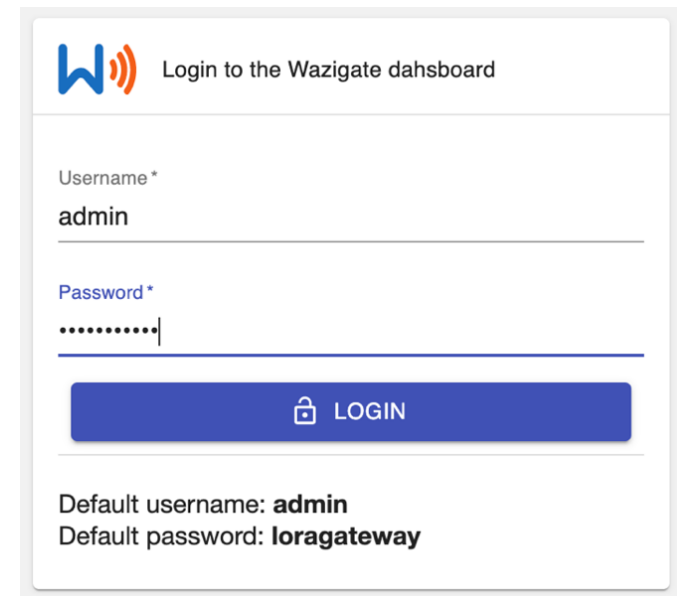


# Checking the gateway (WiFi)

- ④ Use a smartphone to check access to gateway through WiFi
- ④ Connect to WAZIGATE\_XXXXXXXXXXXXX WiFi network
  - ④ default WiFi password is loragateway
- ④ Open web navigator. Go to <http://wazigate.local> or <http://10.42.0.1>



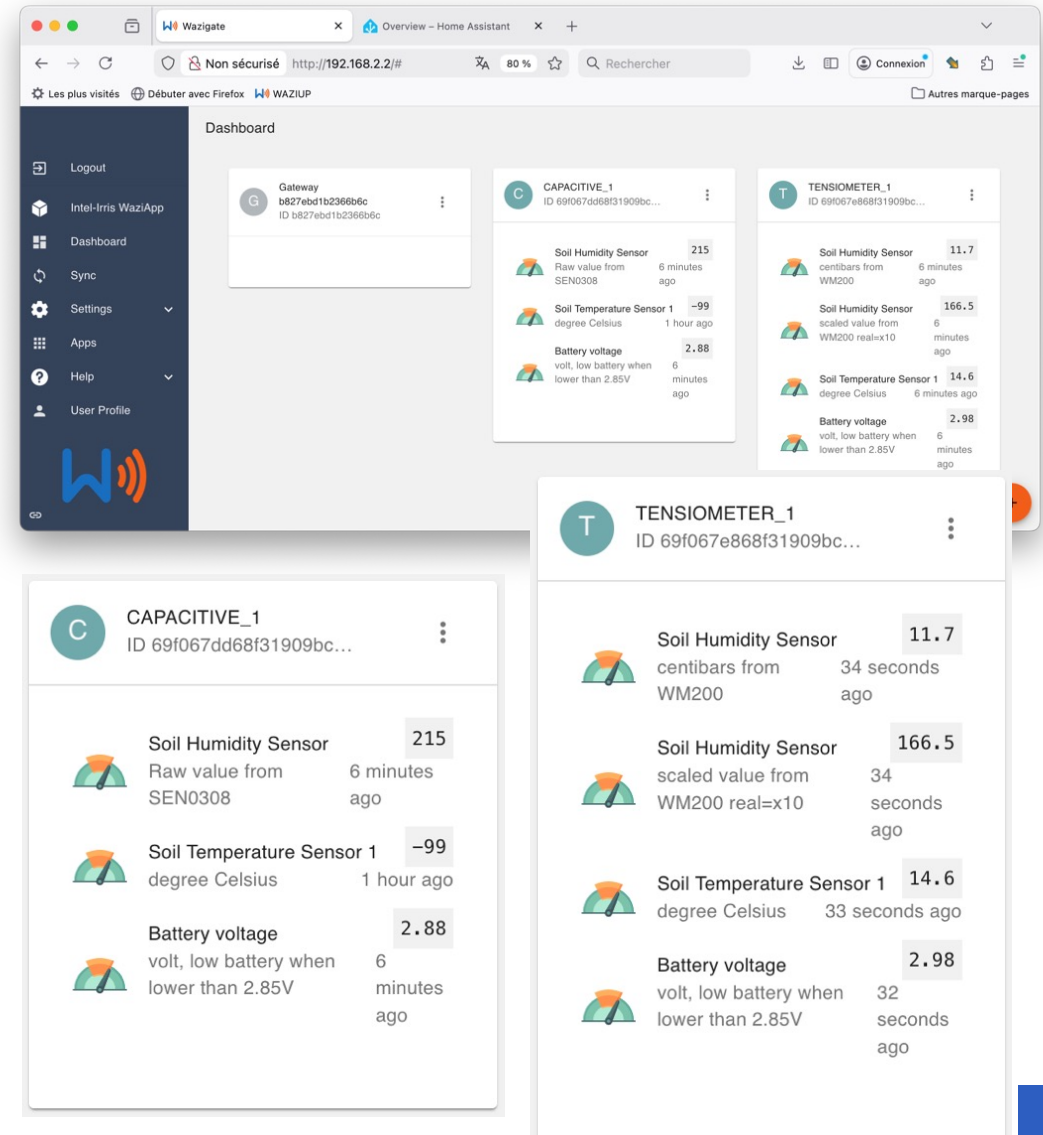
- ④ Use default login to connect
  - ④ User: admin
  - ④ Password: loragateway





# Default gateway configuration

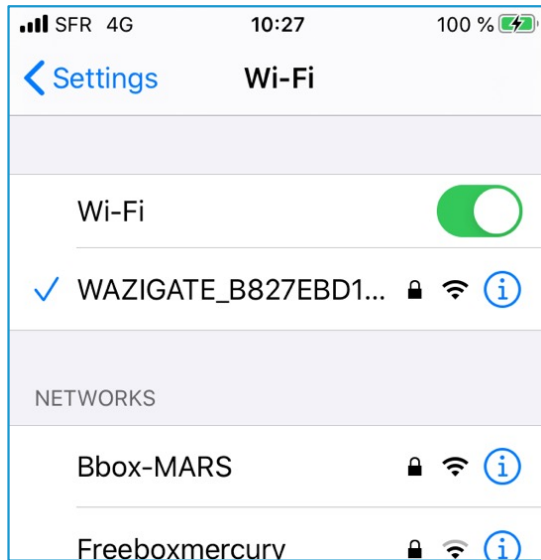
- ④ The default configuration will be ready for
  - ④ 1 capacitive sensor named CAPACITIVE\_1 with address 26011DAA
  - ④ 1 tensiometer sensor named Tensiometer\_1 with address 26011DB1
- ④ Capacitive device will show humidity, soil temperature (if any) and battery level values
- ④ Tensiometer device will show centibar, raw resistance, soil temperature and battery level values







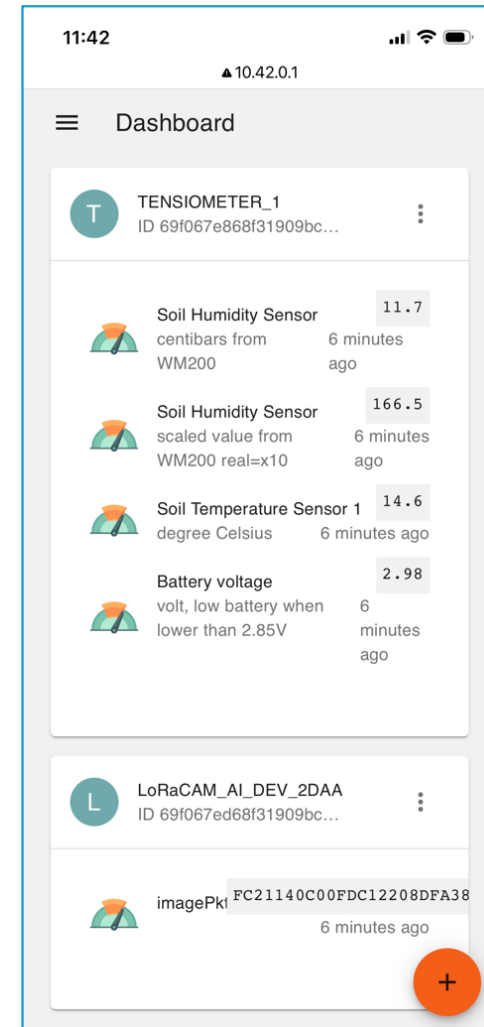
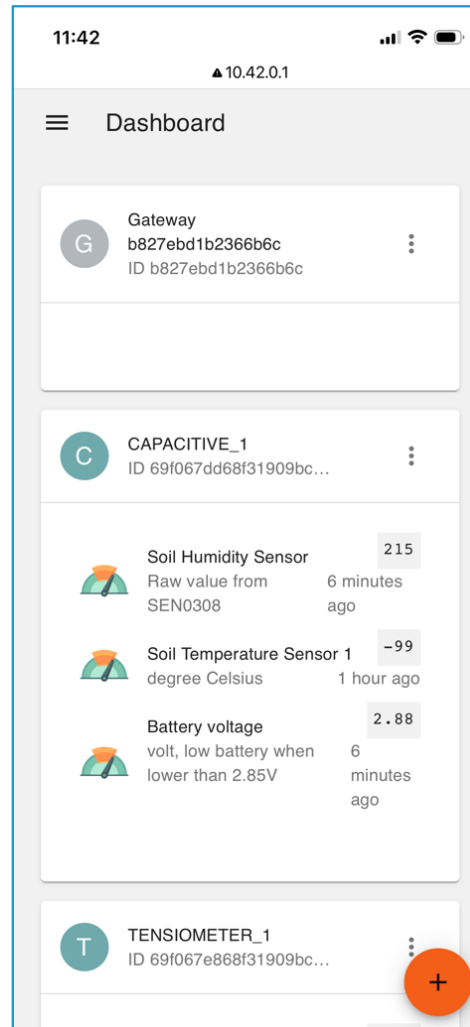
# From a smartphone



<http://wazigate.local>



<http://10.42.0.1>





# Sensor information on OLED

- ⦿ The OLED displays the latest received sensor data for the end-user: the device name, the time of last received data, the sensor raw value and the soil condition
- ⦿ The main screen is displayed for 6s every 30s. Then a screen saver display will show a shorter version of these information with a 5-bar visual
- ⦿ 5 bars: saturated | 4 bars: wet
- ⦿ 3 bars: wet | 2 bars: dry
- ⦿ 1 bar: dry | 0 bar: very dry

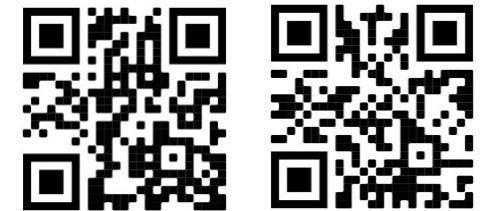
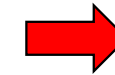






# QR code for connecting to WiFi

- The gateway WiFi is WAZIGATE\_XXXXXXXXXXXX where XXXXXXXXXXXXXXX will be the MAC address of the RPI
- For instance, WAZIGATE\_B827EBD1B236
- A QR code for joining the WiFi network is displayed for 10s before the main screen to join with a smartphone
- Then, users can scan static QR code to connect to the gateway's dashboard
- OLED cycling pattern



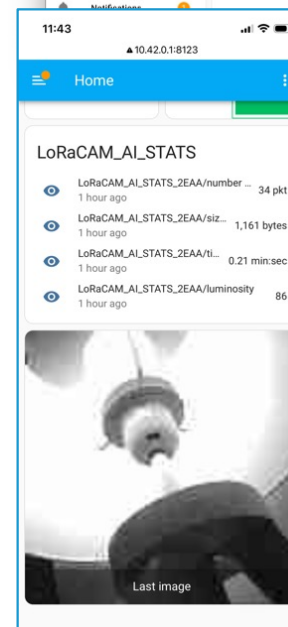
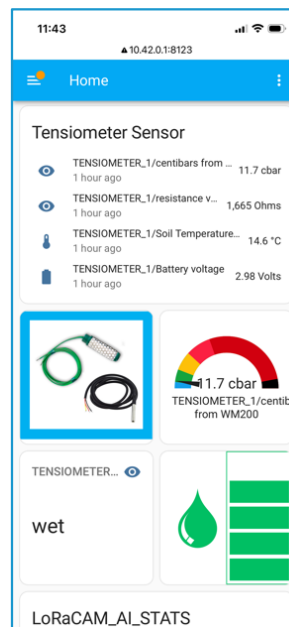
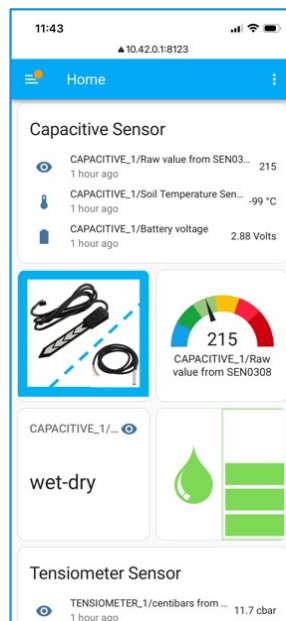
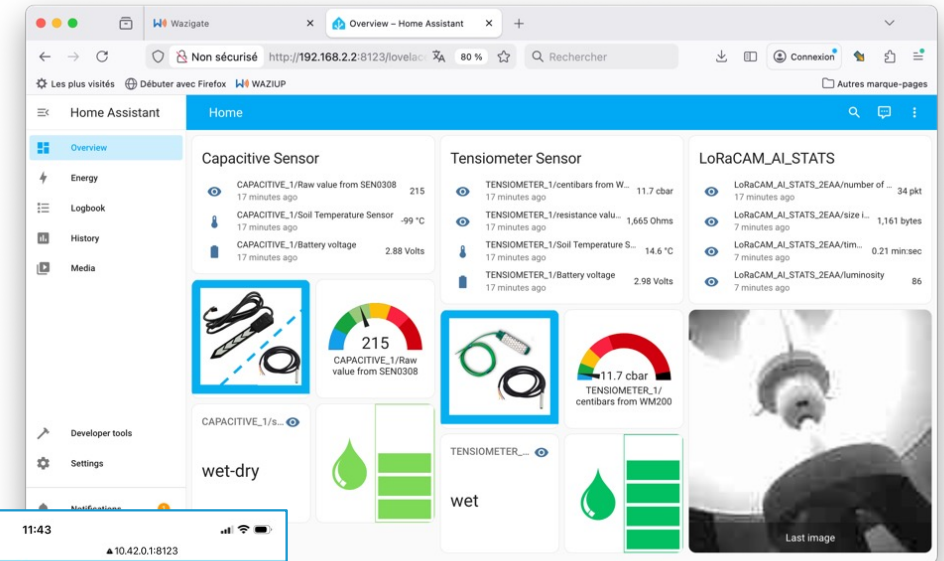
<http://wazigate.local> <http://10.42.0.1>





# Home Assistant dashboard

- Connect to
  - <http://wazigate.local:8123> (Ethernet)
  - <http://10.42.0.1:8123> (gateway's WiFi)
- The HA login page should appear
  - User: intelirris
  - Password: intelirris





# Connect to ChirpStack Network Server

- The generic WaziGate comes with ChirpStack Network Server
- Open web navigator. Go to <http://wazigate.local:8080> or use IP address

ChirpStack Login

Username / email \*

admin

Password \*

\*\*\*\*\*

LOGIN

- Use default login to connect
  - User: admin
  - Password: admin
- Then get access to
  - registered devices
  - live LoRaWAN packet capture
  - downlink, ...

ChirpStack Application Server

Non sécurisé http://192.168.0.32:8080/#/organizations/1/applications/1

ChirpStack

Search organization, application, gateway or device

admin

Applications / Wazigate

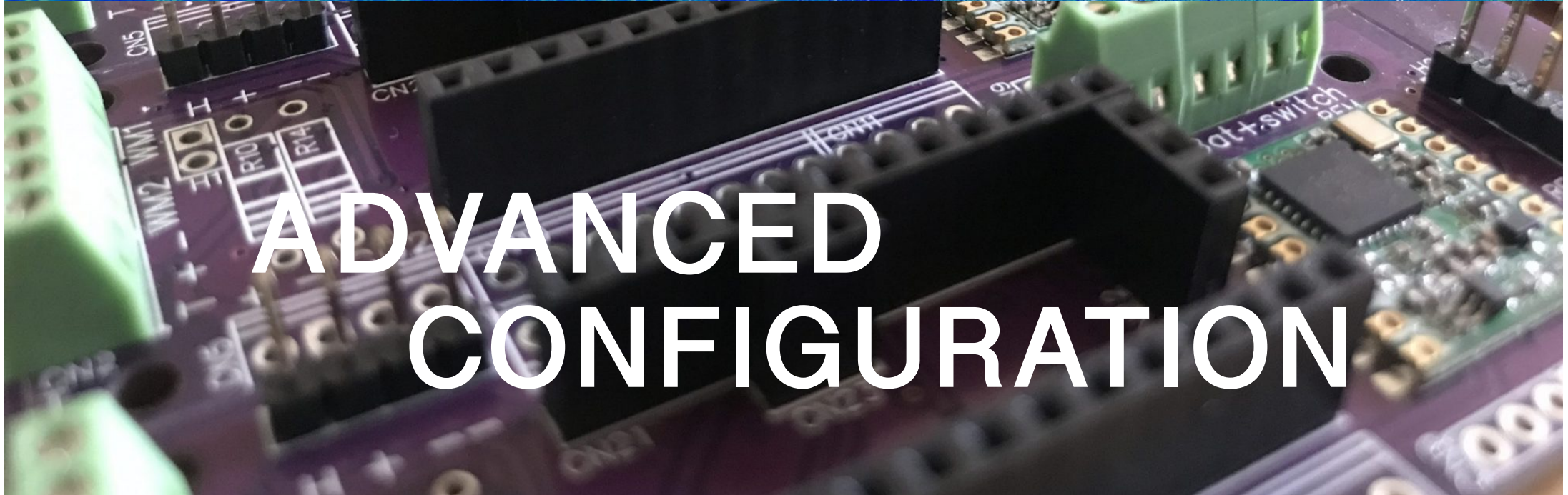
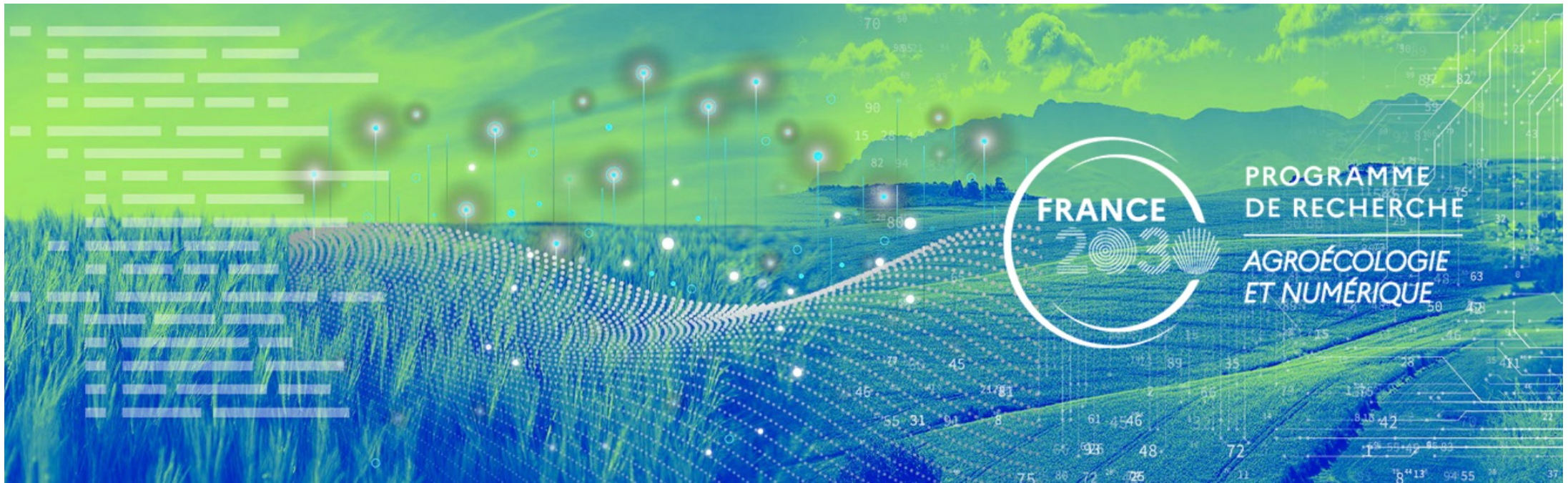
DEVICES APPLICATION CONFIGURATION INTEGRATIONS

+ CREATE

| Last seen  | Device name                      | Device EUI       | Device profile          | Link margin | Battery |
|------------|----------------------------------|------------------|-------------------------|-------------|---------|
| a year ago | <a href="#">AA555A0026011DAA</a> | aa555a0026011daa | <a href="#">Wazidev</a> | n/a         | n/a     |
| n/a        | <a href="#">AA555A0026011DB1</a> | aa555a0026011db1 | <a href="#">Wazidev</a> | n/a         | n/a     |
| a year ago | <a href="#">AA555A0026012DAA</a> | aa555a0026012daa | <a href="#">Wazidev</a> | n/a         | n/a     |
| a year ago | <a href="#">AA555A0026012EAA</a> | aa555a0026012eaa | <a href="#">Wazidev</a> | n/a         | n/a     |

Rows per page: 10 1-4 of 4





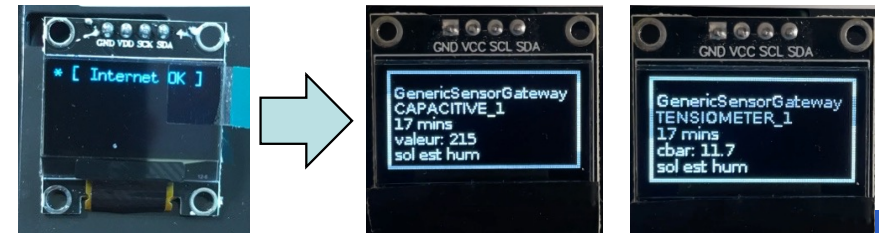




# Syncing RTC module

## Option 1: with Internet, enabling RTC to sync on boot

- Use your laptop that should itself be **connected to the Internet** (using your smartphone's WiFi sharing feature for instance)
- Make sure that your laptop will share its Internet connection to devices connected on its Ethernet port. See how to do so:  
<https://www.waziup.io/documentation/wazigate/v2/install/#connect-with-ethernet-cable-to-pc>
- Before powering gateway, connect it to your laptop by Ethernet cable**
- As the gateway will normally run without Internet once deployed, its clock should be synched with the **RTC that also needs to be synched**
- Then power the gateway. You should see the first [Internet OK] screen. Wait 3-4mins for the main gateway screen to appear
- On boot (and only on boot)**, the RTC module will then be automatically synched with Internet's time & date





# Synching RTC module

## Option 2: without Internet (RTC can not sync on boot)

- It is possible to sync the RTC manually, after boot
- First, power gateway then connect to gateway's WiFi
- Go to Settings then Configuration in Configurations top menu. Set time & date manually. Click on SAVE
- Then use the embedded SSH functionalities (in Settings/Maintenance) to log in the gateway (user pi/ pwd loragateway)
- Then type "date" to check if the previously date & time has been saved
- Then "sudo hwclock -w"
- Check with "sudo hwclock -r"
- It is possible to sync other RTC modules with this method by hot plugging another RTC module

Time Settings

UTC Time : 08:28:43

Local Time : 10:28:43

Date : 2023-07-13

Time Zone : Europe/Paris

Set Timezone:  
Automatic

Set time of the gateway manually:  
13/07/2023 10:27

SAVE

```

wazigate login: pi
Password:
pi@wazigate:~$ date
Thu Jul 13 10:39:36 CEST 2023
pi@wazigate:~$ sudo hwclock -w
pi@wazigate:~$ sudo hwclock -r
2023-07-13 10:39:52.269703+02:00
pi@wazigate:~$
  
```

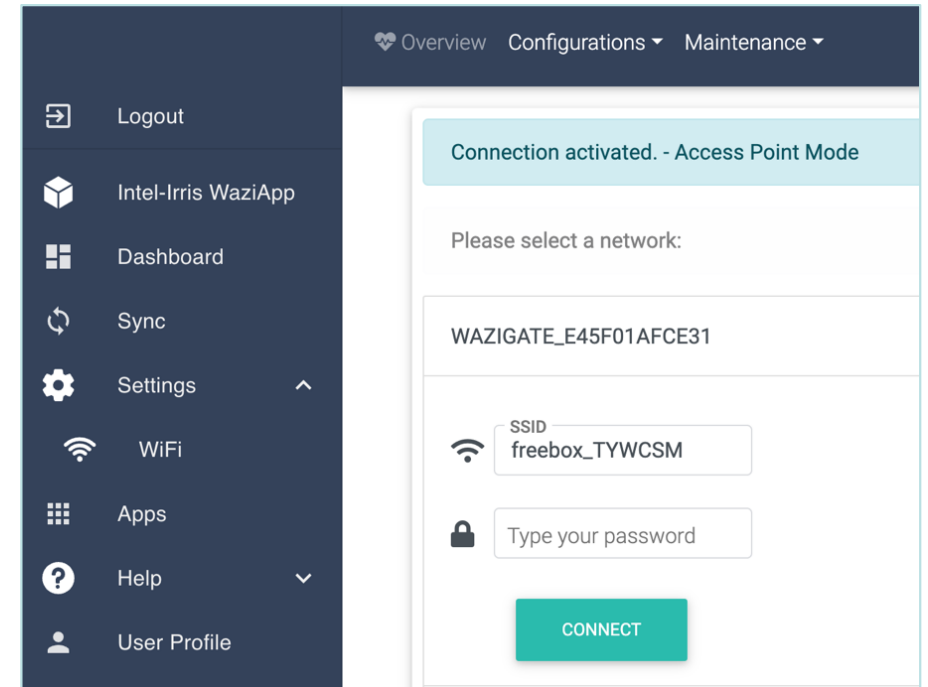




# WiFi client

## connect gateway to a WiFi network

- By default, the gateway acts as a WiFi Access Point
- To connect the gateway to a WiFi network, go to Setting/WiFi to list all available WiFi networks
- Then select the one you want in order to provide the WiFi password
- Ex: connect to freebox\_TYWCSM
- Once connected, gateway is in WiFi Client mode





# WiFi client

connect gateway to a WiFi network, con't

- ⦿ You can connect to several WiFi networks, one after another, to have a list of known WiFi networks
- ⦿ They will be memorized and if the current WiFi network is not available, another available network in the list of known WiFi networks will be selected
- ⦿ If there are no available WiFi networks in the list of known WiFi networks anymore, then the gateway switches back to Access Point mode



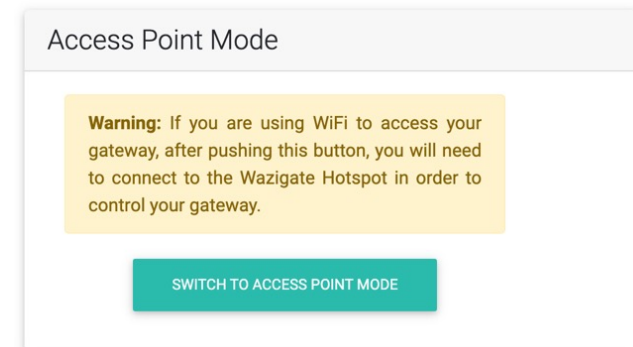
# WiFi AP

## switch gateway back to WiFi access point mode

- ⦿ To get back to Access Point mode, got to Setting/WiFi and simply click on "Forget" for the current WiFi network



- ⦿ If you previously added several WiFi networks, click on "Forget" for **ALL** known & memorized WiFi networks
- ⦿ **DO NOT USE** the "SWITCH TO ACCESS POINT MODE" option
- ⦿ **IT IS NOT STABLE!**



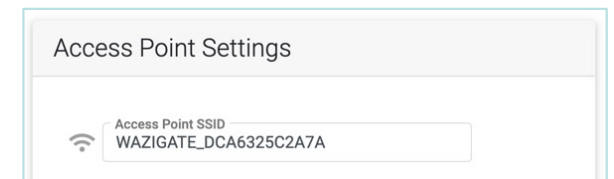
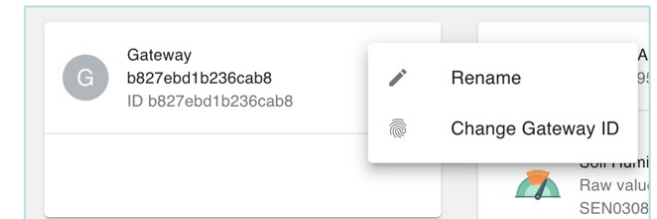




# WaziCloud

## Synching your gateway to the cloud

- If you want to sync your gateway to the WAZIUP Cloud (WaziCloud), look at this tutorial page
  - <https://www.waziup.io/documentation/wazigate/v2/install/#registration-with-the-cloud>
- You will need an account on WaziCloud
  - If you don't have one, you need to create one first
  - <https://dashboard.waziup.io/>
- Then, you **NEED** to change your gateway id
  - Use the unique MAC address of your gateway that appears in Settings/Configuration (it is used as your gateway's WiFi hotspot)
  - Here: DCA6325C2A7A
  - Add 0000 at the end to have 16 digits
  - -> DCA6325C2A7A0000

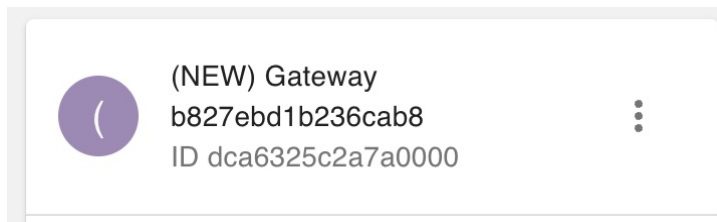




# WaziCloud

## Synching your gateway to the cloud, con't

- ⦿ You should have a new gateway on your dashboard with the new ID



- ⦿ Enter your WaziCloud credentials in the Sync menu
- ⦿ Then, just activate sync on your gateway which needs to be connected to Internet
- ⦿ Log in the WaziCloud dashboard and check that you see your gateway and your dev
- ⦿ You can activate/deactivate synchronization at anytime



# MQTT

## MQTT integration from WaziCloud

- ⦿ With sensor data on WaziCloud, it is possible to subscribe to those data with MQTT protocol
- ⦿ With command line `mosquitto_sub`
  - ⦿ `mosquitto_sub`  
-L "mqtt://api.waziup.io/devices/<deviceId>/sensors/<sensorId>/value"
  - ⦿ `mosquitto_sub`  
-h api.waziup.io -t devices/<deviceId>/sensors/<sensorId>/value
- ⦿ With other MQTT integration client/platform
  - ⦿ Host: api.waziup.io
  - ⦿ Topic: devices/<deviceId>/sensors/<sensorId>/value
- ⦿ Output
  - ⦿ `{"value":34,"timestamp":"2022-12-23T10:23:54Z"}`





# MQTT

## MQTT integration – MQTT client on a smartphone

- Example with an MQTT client (EasyMQTT) on an iPhone7

